

Homework 1

Code ▼

PSTAT 131/231

Machine Learning Main Ideas

Please answer the following questions. Be sure that your solutions are clearly marked and that your document is neatly formatted.

You don't have to rephrase everything in your own words, but if you quote directly, you should cite whatever materials you use (this can be as simple as "from the lecture/page # of book").

Question 1:

Define supervised and unsupervised learning. What are the difference(s) between them?

Question 2:

Explain the difference between a regression model and a classification model, specifically in the context of machine learning.

Question 3:

Name two commonly used metrics for regression ML problems. Name two commonly used metrics for classification ML problems.

Question 4:

As discussed, statistical models can be used for different purposes. These purposes can generally be classified into the following three categories. Provide a brief description of each.

- Descriptive models:
- Inferential models:
- Predictive models:

Question 5:

Predictive models are frequently used in machine learning, and they can usually be described as either mechanistic or empirically-driven. Answer the following questions.

- Define mechanistic. Define empirically-driven. How do these model types differ? How are they similar?
- In general, is a mechanistic or empirically-driven model easier to understand? Explain your choice.
- Describe how the bias-variance tradeoff is related to the use of mechanistic or empirically-driven models.

Question 6:

A political candidate's campaign has collected some detailed voter history data from their constituents. The campaign is interested in two questions:

- Given a voter's profile/data, how likely is it that they will vote in favor of the candidate?
- How would a voter's likelihood of support for the candidate change if they had personal contact with the candidate?

Classify each question as either predictive or inferential. Explain your reasoning for each.

Exploratory Data Analysis

This section will ask you to complete several exercises. For this homework assignment, we'll be working with the `mpg` data set that is loaded when you load the tidyverse. Make sure you load the tidyverse and any other packages you need.

Exploratory data analysis (or EDA) is not based on a specific set of rules or formulas. It is more of a state of curiosity about data. It's an iterative process of:

- generating questions about data
- visualize and transform your data as necessary to get answers
- use what you learned to generate more questions

A couple questions are always useful when you start out. These are "what variation occurs within the variables," and "what covariation occurs between the variables."

You should use the tidyverse and `ggplot2` for these exercises.

Exercise 1:

We are interested in highway miles per gallon, or the `hwy` variable. Create a histogram of this variable. Describe what you see/learn.

Exercise 2:

Create a scatterplot. Put `hwy` on the x-axis and `cty` on the y-axis. Describe what you notice. Is there a relationship between `hwy` and `cty`? What does this mean?

Exercise 3:

Make a bar plot of `manufacturer`. Flip it so that the manufacturers are on the y-axis. Order the bars by height. Which manufacturer produced the most cars? Which produced the least?

Exercise 4:

Make a box plot of `hwy`, grouped by `cyl`. Use `geom_jitter()` and the `alpha` argument to add points to the plot.

Describe what you see. Is there a relationship between `hwy` and `cyl`? What do you notice?

Exercise 5:

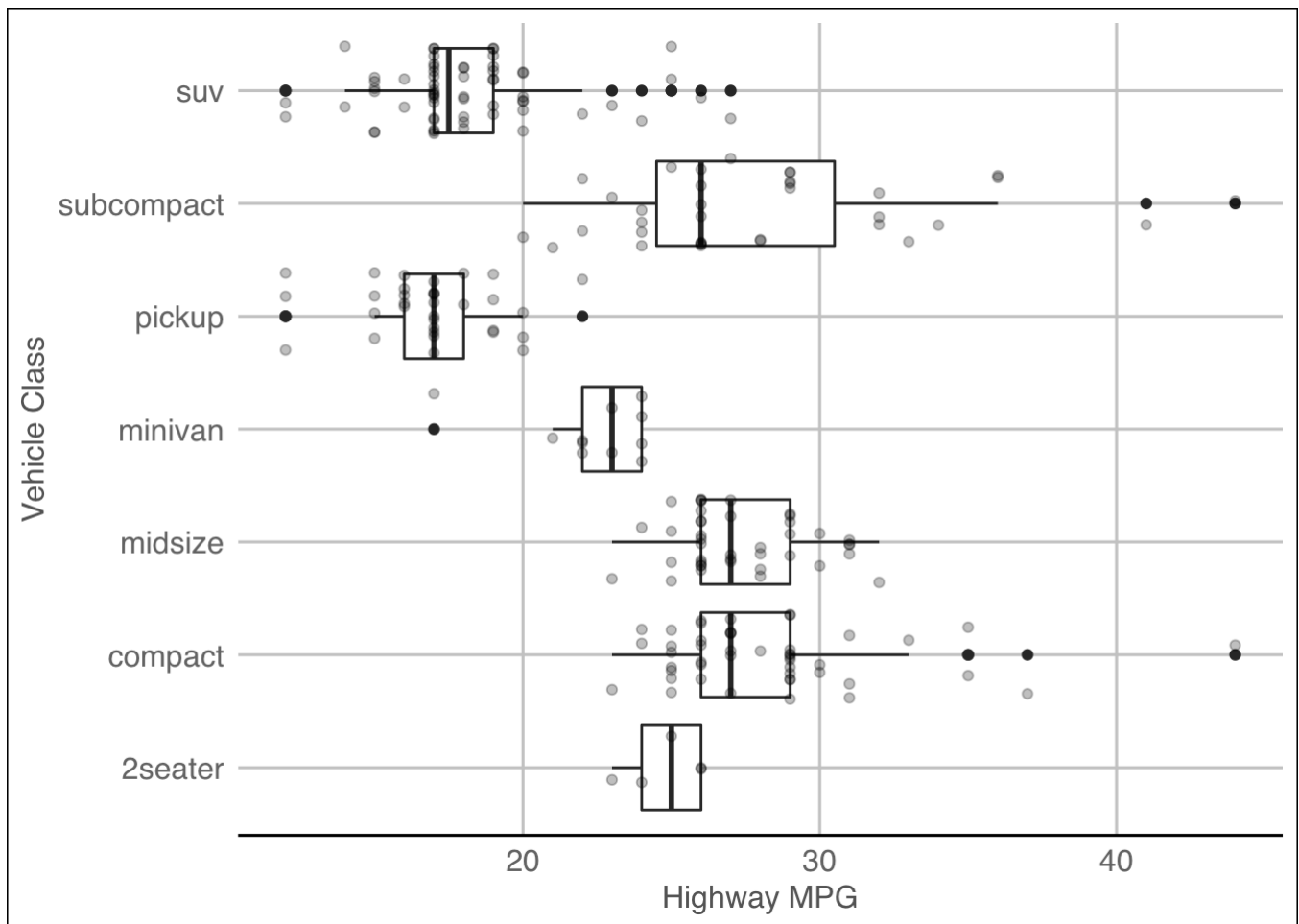
Use the `corrplot` package to make a lower triangle correlation matrix of the `mpg` dataset. (Hint: You can find information on the package here. (<https://cran.r-project.org/web/packages/corrplot/vignettes/corrplot-intro.html>))

Which variables are positively or negatively correlated with which others? Do these relationships make sense to you? Are there any that surprise you?

231 Students Only:

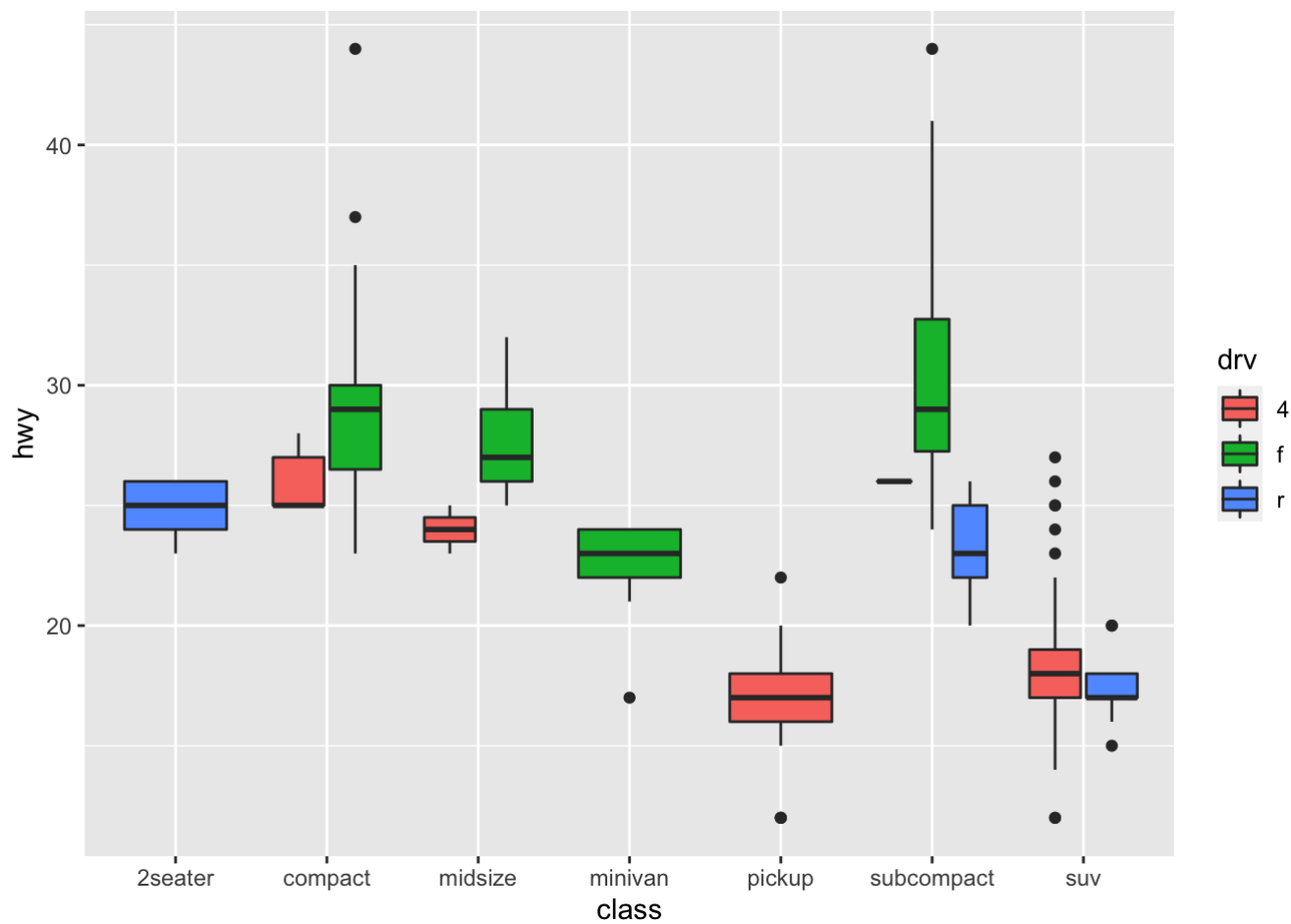
Exercise 6:

USing `ggplot()` , recreate the following graphic as closely as you can. *Hint: Use the `ggthemes` package.*



Exercise 7

Recreate the following graphic.



Exercise 8

Recreate the following graphic.

